

Sazzad Hossen

Dhaka, Bangladesh

B.Sc. in Electrical & Electronic Engineering

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LinkedIn Profile

Sazzad Hossen

RESEARCH INTERESTS

Analog and mixed-signal integrated circuit design, RF circuit design, VLSI layout, low-power CMOS design.

EDUCATION

B.Sc. in Electrical & Electronic Engineering (Major: Electronics, Minor: Communication)	Feb 2022 – Feb 2026
United International University (UIIU)	CGPA: 3.92 / 4.00
Higher Secondary Certificate (HSC)	2020
Government Science College	GPA: 5.00 / 5.00
Secondary School Certificate (SSC)	2018
Shiruail High School	GPA: 5.00 / 5.00

WORK EXPERIENCE

Analog IC Design Intern	Jul 2026 – Present
Silicon Array	Dhaka, Bangladesh
– Design and optimize transistor-level analog IC schematics and run circuit simulations. – Implement custom physical layouts with a focus on floorplanning, matching, and symmetry. – Execute DRC and LVS checks to verify layout-to-schematic accuracy.	
Teaching Assistant	Mar 2026 – Jul 2026
United International University (Dept. of EEE)	Dhaka, Bangladesh
– Conducted lab demonstrations, led tutorials, and provided student counseling. – Assisted faculty with course management and exam invigilation.	
Undergraduate Assistant	Mar 2025 – Mar 2026
United International University (Dept. of EEE)	Dhaka, Bangladesh
– Assisted faculty during lab classes and conducted extra tutorial sessions for difficult coursework. – Provided student academic counseling and supported BAETE program accreditation activities.	
Undergraduate Grader	2022 – 2024
United International University (Dept. of EEE)	Dhaka, Bangladesh
– Evaluated assignments and lab reports for multiple EEE courses. – Maintained grading records and performance analysis using Excel.	

PUBLICATIONS

- **S. Hossen**, S. Karmakar, T. Hossain, A. H. Saykat, R. Mostafa, and M. Hasanuzzaman, "Design and Performance Analysis of a 2.45 GHz Low-Power RF Receiver Using 45nm CMOS Technology in Cadence Virtuoso," presented at IEEE MARSCON 2026, Dhaka, Bangladesh, Aug. 2026. (To appear in IEEE Xplore)

CAPSTONE/THESIS PROJECT

- **Design of a 2.45 GHz RF Receiver in CMOS 45nm technology** : Designed and simulated a complete RF receiver chain by seamlessly integrating BPF, LNA, Mixer, and LPF blocks to optimize signal processing and minimize system noise. Achieved an overall conversion gain of 50 dB with a down-converted intermediate frequency (IF) of 10–20 MHz

ACADEMIC PROJECTS

- **Design of Schematic and Layout of 8-bit Carry Save Adder** : Completed schematic simulation, functional verification, and layout implementation ensuring DRC & LVS clean.
- **Design of 2-Stage OTA in 45nm CMOS Technology**: Designed and simulated a high-gain, wide-bandwidth OTA using CMOS 45nm technology.
- **Fully Differential Folded-Cascode OTA**: Designed in 45nm CMOS using gm/ID and verified in Virtuoso.
- **32-bit ARM ALU (Verilog)**: Developed RTL code for a 32-bit ALU featuring arithmetic, logic, and shift operations with full NZCV flag logic.

Other Projects: Smart Glove for Sign Language Translation | BPSK Modulation/Demodulation (Simulink and Hardware) | DSB-SC Signal Transmission & SDR Power Measurement | License Plate Recognition (MATLAB) |

SKILLS

Programming: C, Verilog, MATLAB

EDA Tools: Cadence Virtuoso, NCSim, ETAP

Simulation & Design: Proteus, Multisim, AutoCAD

Hardware: STM32, Arduino Uno

Soft Skills: Communication & Presentation, Leadership, Teaching & Mentoring

ACHIEVEMENTS

Champion VLSI Design Competition (EEE Day 2025)

Champion Junior Intra-University VLSI Design Competition

Champion MATLAB: Simulate & Innovate

Runner-Up VLSI Design Competition, MIST Robolution

Runner-Up EEE Project Show (Senior Category)

Scholarship University Award for Academic Excellence Scholarship — Recipient for 12 consecutive trimesters

PROFESSIONAL DEVELOPMENT

Workshops & Training: Professional Skills Development, Workshop on Basic VLSI Design, MATLAB: Simulate & Innovate, Foundations of Real-World Mathematics, Public Speaking & Presentation Skills, Introduction to Computer Networks, DevOps for Engineers.

MEMBERSHIPS

UIU Electrical & Electronics Club, Govt. Science College Science Club, Shiruail High School Science Club.

REFERENCES

Dr. Raqibul Mostofa

Professor & Dean, School of Science & Eng.

Director, Biomedical, Image & Signal Research

United International University, Dhaka

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Relation: Undergraduate Capstone Project Co-Supervisor

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